Sathvik Vadavatha

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Professional Summary

Software and AI engineer with 3 years of experience designing and deploying end to end AI systems across fintech and healthcare, spanning full stack application development, agentic workflows, cloud infrastructure, and ML model integration. Experienced taking systems from data layer through model deployment to production facing applications, with hands on work across AWS, Azure, and GCP in regulated and enterprise environments. Holds Databricks Certified Data Engineer Professional and AWS Certified Machine Learning Engineer certifications.

Education

Master of Science, Computer Information Systems Jan 2024 – Dec 2025
Northeastern University, Boston, MA

Bachelor of Science, Mechanical Engineering Aug 2018 – Apr 2022
University of Illinois at Urbana-Champaign, IL

Certifications

Databricks Certified Data Engineer Professional (Credential ID 183309586)
AWS Certified Machine Learning Engineer, Associate

Professional Experience

Stratafi (Software and AI Engineer) | Boston, MA May 2026 – Present

- Built agentic AI systems supporting Stratafi's equipment financing underwriting platform, automating equipment identification and data enrichment to speed up credit decisioning, cutting manual equipment review time by ∼35% for loan officers evaluating equipment backed loans.
- Developed an AI driven agent browser that autonomously navigates auction and marketplace sites to extract structured equipment data (make, model, specs, images, price) from a listing URL or lot number, reducing per listing data entry time from ∼10 minutes to under 30 seconds.
- Integrated Plaid for real time bank account and cash flow verification and Enigma for business bankruptcy and risk checks into Stratafi's credit engine, cutting borrower risk assessment turnaround time by ∼40% within the underwriting pipeline.
- Extended Postgres schemas mapping independently sourced equipment market data onto Stratafi's existing classification taxonomy across 80+ categories and 100K+ models, enabling scalable comparison against third party valuation benchmarks.

Arrivia Inc (Data & AI Engineer) | Contract | Boston, MA Jan 2026 – May 2026

- Consolidated 5+ third party travel platforms across booking, rewards, and behavioral data into a unified Azure Data Lake supporting an agentic travel recommendation platform for 2M+ loyalty members, eliminating data silos and enabling real time agent reasoning.
- Engineered real time data ingestion and feature services on Azure Databricks feeding the recommendation platform, improving recommendation relevance for product and marketing teams.
- Designed event driven workflows transforming loyalty and transaction patterns into structured ML feature sets, cutting feature preparation time by ∼40% for downstream recommendation and targeting models.
- Optimized distributed Python services through parallelism tuning and memory configuration, reducing pipeline execution time by ∼30% and improving feature freshness for 10K+ daily active users.
- Automated CI/CD pipelines on Azure DevOps for deployment of data services and notebooks across dev and production, cutting manual deployment time by ∼50%.

Triple Ring Technologies (ML Engineer Intern) | Boston, MA Jun 2025 – Sep 2025

- Established a new ML platform from the ground up on AWS using Terraform and Step Functions for FDA regulated medical device software, provisioning compliant dev/staging/prod environments from scratch and reducing setup time from days to under an hour.
- Delivered end to end SageMaker training and inference pipelines for Random Forest and U-Net CNN models with MLflow tracking across 3 model versions, cutting deployment time by ∼60% versus manual processes.
- Orchestrated model workflows on AWS Step Functions across 6 pipeline stages, eliminating manual handoffs and ensuring full traceability for FDA audit submissions.
- Constructed a scalable data registry on AWS Glue and S3 organizing medical data across raw, curated, and feature layers, cutting ML engineer discovery time by ∼50% and ensuring versioned inputs across all runs.
- Configured SageMaker model monitoring to track inference performance and data drift across deployed models, catching degradation before it reached clinical workflows.

ITC Infotech USA (Data Engineer) | Frederick, MD Apr 2022 – Dec 2023

- Refactored Python and PySpark services extracting product and engineering change data from Baker Hughes' legacy PLM system (Windchill) across 6 product domains, increasing daily throughput from ∼1,500 unreliable records to 3,000–5,000 stable records in Snowflake.
- Re-architected batch processing onto Databricks with Delta Lake, introducing incremental ingestion and audit ready versioning and reducing pipeline failure rate by ∼70%.
- Streamlined event driven Kafka and PySpark Structured Streaming services capturing real time product change events from Windchill, reducing data latency from 8+ hours to under 5 minutes for engineering and operations teams.
- Standardized Snowflake schemas across raw, curated, and reporting tiers covering 10+ analytical tables, unifying KPIs for product lifecycle and revision history consumed by engineering and supply chain teams.
- Enforced schema contracts via Airflow based validation logic before warehouse loads, diagnosing and resolving 40+ production failures and cutting recurring incidents by ∼65% over 6 months.

Projects

Sanq.ai (Founder) | *FastAPI, Supabase (PostgreSQL), Expo React Native* | sanq.ai

- Founded and built a multi tenant restaurant inventory management platform used by a live pilot restaurant; the platform uses AI to read photos of invoices and menus, automatically match items to inventory, and flag anything it is not confident about instead of guessing, letting staff track stock by sending a photo instead of manual logging.

CitadelTwin | *React, TypeScript, FastAPI, YOLO, EasyOCR, Shapely, trimesh*

- Built a platform that converts 2D architectural blueprints into interactive 3D building models in seconds instead of hours; the underlying pipeline detects floors and rooms, labels them correctly, and validates that the layout makes physical sense, reaching 89% accuracy on wall detection.

TxSentry | *LightGBM, LangGraph, PySpark, Kafka, FastAPI* | GitHub

- Built a fraud detection system for financial transactions; the system combines a machine learning model (99.4% accuracy distinguishing fraud from legitimate transactions) with an AI agent that investigates flagged transactions using 11 specialized tools and generates a case summary, reducing manual review work for fraud analysts.

NutriGuard AI | *FastAPI, GPT-4, Pinecone, React Native* | GitHub

- Built a mobile app that scans food nutrition labels and tells users whether a product fits their dietary needs; the app cross references scanned ingredients against a database of 20,000+ FDA food records and uses GPT-4 to generate personalized recommendations grounded in that verified data rather than the AI making things up.

Technical Skills

Programming: Python, SQL, PySpark
Frameworks & Libraries: FastAPI, LangChain, LangGraph, Pandas, NumPy, NetworkX, HuggingFace
ML & AI: Scikit-learn, PyTorch, MLflow, SHAP, OpenAI/Anthropic APIs, RAG, Agentic AI Systems, Prompt Engineering, Claude Code
Data & Streaming: Apache Spark, Kafka, Databricks, Delta Lake, Airflow, Snowflake, dbt
Cloud Platforms: AWS (S3, Glue, Lambda, EC2, SageMaker, Step Functions, Athena, ECS Fargate, Secrets Manager), Azure (Data Factory, Synapse, Databricks, Blob Storage), GCP (BigQuery, Cloud Storage)
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Supabase, Pinecone, Weaviate
Frontend: Next.js, React, TypeScript, Tailwind CSS
DevOps & Tools: Terraform, Docker, GitHub Actions, Azure DevOps, CI/CD, Git
Methodologies: Agile/Scrum, HIPAA, FDA Design Controls, Data Governance